

# Food Frequency Patterns and Nutrient Deficits among Urban Slum Children in Varanasi, Uttar Pradesh: Evidence from a Community-Based Dietary Assessment Study

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## ABSTRACT

Nutritional deprivation among children residing in urban slums continues to be a major public health concern in India. Dietary inadequacies, infrequent meals, low dietary diversity, and limited access to nutrient-rich foods contribute significantly to undernutrition and micronutrient deficiencies. This study assessed food frequency patterns, dietary habits, dietary diversity, and associated nutrient deficits among school-age children living in urban slums in Varanasi, Uttar Pradesh. We conducted a community-based cross-sectional study in 350 children aged 5–12 years from six selected slums of Varanasi. We assessed dietary intake using a validated Food Frequency Questionnaire (FFQ) and the 24-hour dietary recall method. Nutrient intake was estimated using Indian Food Composition Tables and compared with Recommended Dietary Allowances (RDA) prescribed by the Indian Council of Medical Research (ICMR). Anthropometric assessment used WHO BMI-for-age standards, Gomez classification, and Waterlow classification. Results show that most children (95.7%) belonged to non-vegetarian households; however, 75.1% consumed only two meals daily. All children frequently consumed rice and roots/tubers (100%), followed by pulses (70.6%), vegetables (71.1%), and wheat (65.1%). Consumption of green leafy vegetables (1.7%), milk products (16.9%), and fruits (0%) was extremely low. Fast-food consumption was remarkably high, with frequent intake of biscuits/samosa (83.7%), namkeen (72%), Kurkure (63.4%), chips (58.9%), and panipuri (52.9%). This study concludes that slum children's diets are characterised by poor dietary diversity and inadequate consumption of nutrient-rich foods, increasing the risk of chronic undernutrition and micronutrient deficiencies. Strengthening nutrition-sensitive interventions targeting dietary diversification and meal quality is urgently needed.

**Keywords:** Food Frequency Questionnaire, Dietary Diversity, Nutritional Deficiency, Urban Slums, School-Age Children, Varanasi

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## INTRODUCTION

Childhood nutrition is a critical determinant of growth, cognitive development, educational achievement, and long-term health outcomes. School-age children require adequate energy and nutrients to support rapid growth and increasing physiological demands. However, children residing in urban slums frequently experience nutritional disadvantages arising from poverty, overcrowding, food insecurity, inadequate sanitation, and poor access to health services. [1,11-13]

India is still carrying one of the world's largest burdens of childhood undernutrition despite significant economic growth. According to national surveys, urban slum populations remain disproportionately affected by malnutrition because of limited dietary diversity and poor food purchasing power. Dietary quality is increasingly recognised as a stronger predictor of nutritional status than calorie intake alone. [3,4,13]

Food Frequency Questionnaires (FFQs) are widely

used to assess habitual food intake and dietary diversity in population-based nutritional studies. FFQs help identify food-consumption patterns that contribute to nutrient inadequacies and adverse health outcomes. [6-8]

Urban slum households often rely heavily on inexpensive cereals and starchy foods while consuming limited amounts of fruits, vegetables, milk products, and animal-source foods. Such dietary patterns increase the risk of deficiencies of iron, calcium, folate, vitamin A, vitamin C, and other essential micronutrients. [5,13,15]

Varanasi is most densely populated and one of the oldest cities in India and has witnessed substantial growth in urban slum settlements. Despite numerous government nutrition programs, evidence on food consumption patterns among school-age slum children remains limited. Therefore, this study assessed food frequency patterns and dietary nutrient deficits among children aged 5–12 years residing in selected slums of Varanasi. [3]

## 1. Materials and Methods

### 1.1 Study Design

This community-based cross-sectional study was conducted among urban slum children in Varanasi. The study formed part of a larger investigation assessing nutritional status and cognitive development among slum children.

### 1.2 Study Area

The study was conducted in six urban slums of Varanasi:

- Durgakund
- Sunderpur
- Nagwa
- Chhittupur
- Bhawaniya Pokhari
- Shankuldhar

These slums were selected from Varanasi Municipal Corporation records.

### 1.3 Sample Size and Participants

The study included 350 children aged 5–12 years, following sample size estimation procedures that accounted for cluster design and expected attrition.

#### Inclusion Criteria

- Children aged 5–12 years
- Residents of selected slum areas
- Caregivers willing to provide informed consent

#### Exclusion Criteria

- Children below 5 years
- Children with diagnosed neurological disorders
- Children with congenital heart disease or chronic illnesses affecting nutritional status

### 1.4 Dietary Assessment

Dietary assessment was done by using a food frequency questionnaire and 24-hour diet recall. [6-

**Table 1. Food Habits of Study Participants (n=350)**

| Food Habit     | Frequency | Percentage |
|----------------|-----------|------------|
| Vegetarian     | 9         | 2.6        |
| Non-vegetarian | 335       | 95.7       |
| Eggitarian     | 6         | 1.7        |

**Table 2. Daily Meal Pattern**

8]

### Food Frequency Questionnaire

Food consumption frequency was recorded for major food groups, including Cereals, Pulses, Green leafy vegetables, other vegetables, Fruits, Milk and milk products, Meat and poultry, Roots and tubers, Oils and fats, and Sugars and fast foods.

### 24-Hour Dietary Recall

We used the dietary recall method to estimate average daily nutrient intake. Nutrient calculations were performed using Indian Food Composition Tables and DietSoft software. Nutrient adequacy was assessed against ICMR Recommended Dietary Allowances. [2,8]

### 1.5 Anthropometric Assessment

We measured weight, height, BMI, and MUAC according to standard procedures. We classified nutritional status using WHO BMI-for-age standards and the Waterlow classification. [1,16]

### 1.6 Statistical Analysis

Descriptive statistics summarised dietary patterns and nutritional outcomes. Results were presented as frequencies, percentages, and mean values.

**Parental Consent:** Written informed consent was obtained from all mothers/caregivers before enrolling them in the study.

## 3. Results

### 3.1 Food Habits and Meal Patterns

The food habit table shows that most respondents (95.7%) were non-vegetarian, while 2.6% were vegetarian and only 1.7% were eggitarian. Three-fourths of children consumed only two meals daily, indicating widespread breakfast skipping and reduced meal frequency. Such practices may adversely affect daily nutrient adequacy and academic performance. [17,18]

|                        |       |
|------------------------|-------|
| Rice                   | 100.0 |
| Roots & Tubers         | 100.0 |
| Tea                    | 99.4  |
| Other Vegetables       | 71.1  |
| Pulses (Dal)           | 70.6  |
| Wheat                  | 65.1  |
| Milk & Milk Products   | 16.9  |
| Green Leafy Vegetables | 1.7   |
| Fruits                 | 0     |

| Meal Pattern    | Frequency | Percentage |
|-----------------|-----------|------------|
| Two meals/day   | 263       | 75.14      |
| Three meals/day | 87        | 24.86      |

### 3.2 Food Frequency Patterns

Food-frequency analysis showed frequent consumption of rice (100%), roots and tubers

(100%), dal (70.6%), and other vegetables (71.1%). Protective foods such as fruits, green leafy vegetables and milk products were consumed

infrequently. High intake of snack foods including biscuits, chips and packaged foods was observed.

**Table 3. Frequency of Consumption of Major**

**Graph 1. Food Frequency Consumption Pattern**

**3.3 Fast-Food Consumption Pattern**

A major finding was the exceptionally high consumption of commercially available snack foods and street foods among slum children. [14,15] Frequent consumption of biscuits/samosa was reported by 83.7% of children, followed by namkeen

| Food group             | Frequent Consumption (%) |
|------------------------|--------------------------|
| Rice                   | 100.0                    |
| Roots & Tubers         | 100.0                    |
| Tea                    | 99.4                     |
| Other Vegetables       | 71.1                     |
| Pulses (Dal)           | 70.6                     |
| Wheat                  | 65.1                     |
| Milk & Milk Products   | 16.9                     |
| Green Leafy Vegetables | 1.7                      |
| Fruits                 | 0                        |

**Figure 2. Fast-Food Consumption Pattern**

**3.4 Dietary Diversity Assessment**

The dietary diversity pattern clearly demonstrates dependence on cereal-based foods with insufficient

**Table 4. Classification of Dietary Diversity**

| Food Group Category | Consumption Level |
|---------------------|-------------------|
| Cereals             | Very High         |
| Pulses              | Moderate          |
| Vegetables          | Moderate          |

The comparison of healthy food versus fast food clearly demonstrates that energy-dense snack foods

**Figure 3. Healthy Foods versus Fast Foods**

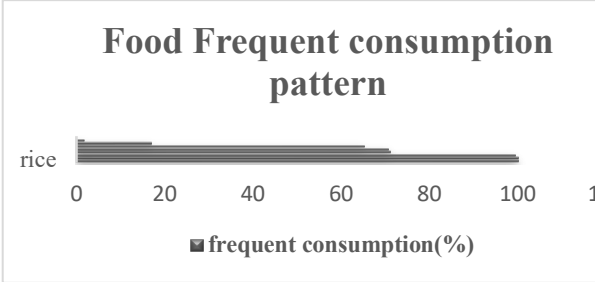
**3.5 Nutritional Implications of Food Frequency Patterns**

The anthropometric findings provide further evidence of the nutritional vulnerability of this population. Only 31.4% of children were classified

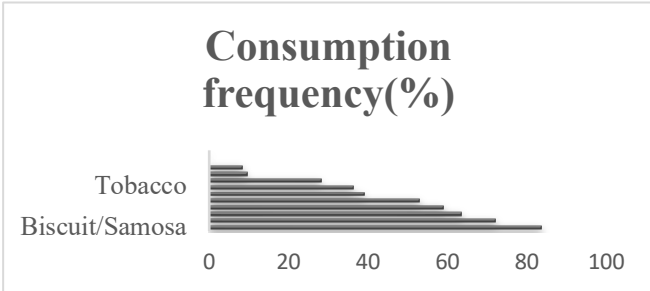
**Table 4. Probable Nutrient Deficits Based on Food Consumption Patterns**

| Category          | Number | Percentage |
|-------------------|--------|------------|
| Normal            | 110    | 31.4       |
| Mild Thinness     | 114    | 32.6       |
| Moderate Thinness | 84     | 24.0       |

**Food Groups**



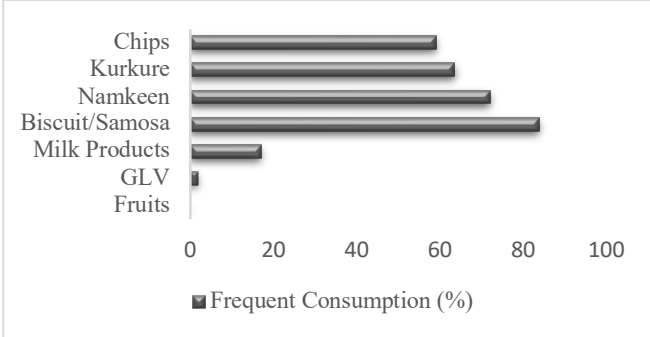
(72%), Kurkure (63.4%), chips (58.9%), Panipuri (52.9%), and aloo chop (38.9%). Chaat was frequently consumed by 28% of respondents. In addition, tobacco consumption was reported by 36% of children.



intake of protective foods such as fruits, milk products, and green leafy vegetables. [9,10,17]

|               |               |
|---------------|---------------|
| Milk Products | Low           |
| Fruits        | Extremely Low |
| GLV           | Extremely Low |
| Fast Foods    | Very high     |

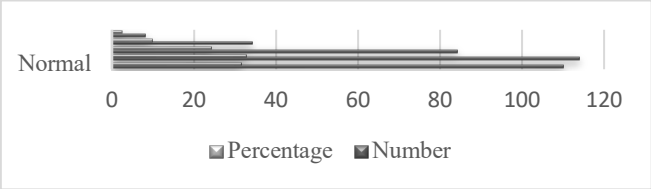
were consumed far more frequently than nutrient-rich foods.



as normal according to BMI-for-age, whereas 32.6% had mild thinness, 24.0% moderate thinness, and 9.7% severe thinness. Thus, 68.3% of the children were classified as thin to some degree, while only 2.3% were overweight. [1,11,16]

|                 |    |     |
|-----------------|----|-----|
| Severe Thinness | 34 | 9.7 |
| Overweight      | 8  | 2.3 |

**Graph 4. BMI-for-Age Classification**



3.6 Stunting and Wasting

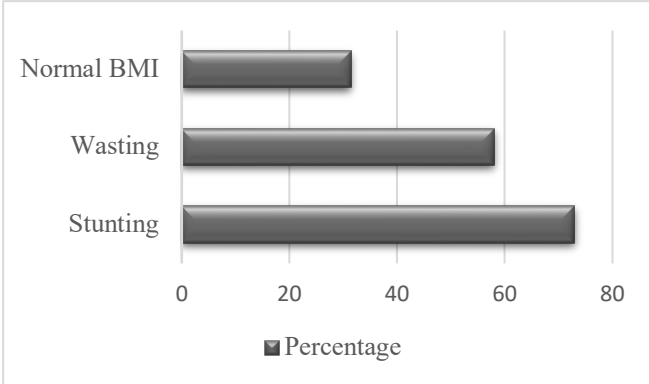
Waterlow classification done on these children shows 72.9% stunting and 58.0% wasting, and only 31.4% had a normal BMI. The high prevalence of

Table 7. Chronic and Acute Undernutrition

| Indicator  | Percentage |
|------------|------------|
| Stunting   | 72.9       |
| Wasting    | 58.0       |
| Normal BMI | 31.4       |

Graph 5. Burden of Undernutrition

stunting (72.9%) and wasting (58.0%) further indicates chronic and acute nutritional problems in the study population.



4. Discussion

The findings of the present study reveal a highly monotonous dietary pattern among urban slum children in Varanasi. Rice, wheat, and roots/tubers constituted the primary components of the daily diet. Such cereal-dominated diets are common among economically disadvantaged populations because cereals provide relatively inexpensive energy compared with fruits, vegetables, and animal-source foods. [13,17,18]

The most concerning finding was the extremely low consumption of fruits and green leafy vegetables. These foods are major sources of vitamins, minerals, antioxidants, and dietary fiber. Their inadequate consumption increases susceptibility to micronutrient deficiencies, impaired immunity, poor growth, and reduced cognitive performance. [5,17,18]

Milk and milk product consumption was also remarkably low. Dairy foods are important sources of calcium, high-quality protein, riboflavin, and vitamin B12. Low intake may contribute to poor bone mineralization and impaired skeletal growth among children. [2,18]

Breakfast skipping emerged as another important nutritional concern. Approximately three-fourths of children consumed only two meals daily. Previous research has shown that breakfast omission is associated with lower micronutrient intake, poorer academic performance, and reduced concentration. [19]

One of the most significant findings of the present study was the coexistence of poor dietary diversity and excessive fast-food consumption. While fruits, green leafy vegetables, and milk products were rarely consumed, processed snack foods such as biscuits, namkeen, Kurkure, chips, and panipuri were consumed frequently by more than half of the study population. [14,15,20]

The dietary inadequacies observed through FFQ assessment were reflected in anthropometric

outcomes. Nearly seven out of ten children were undernourished according to WHO BMI-for-age criteria, while stunting affected almost three-fourths of the study population. These findings indicate long-term nutritional deprivation and chronic dietary insufficiency. [1,11-13,16]

5. Conclusion

The present study demonstrates that urban slum children in Varanasi consume diets characterized by:

- Excessive dependence on cereals and tubers
- Poor dietary diversity
- Extremely low fruit consumption
- Very low intake of green leafy vegetables
- Inadequate milk and dairy consumption
- High prevalence of breakfast skipping

These dietary patterns are associated with widespread undernutrition, stunting, and wasting. Community-based nutrition education, school feeding programs, dietary diversification strategies, and improved accessibility of affordable nutrient-rich foods should be prioritized to improve the nutritional well-being of urban slum children. [5,20]

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